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Particulate matter is solid particles and liquid particles suspended in air. Polluted air contains more particulate matter than clean air.

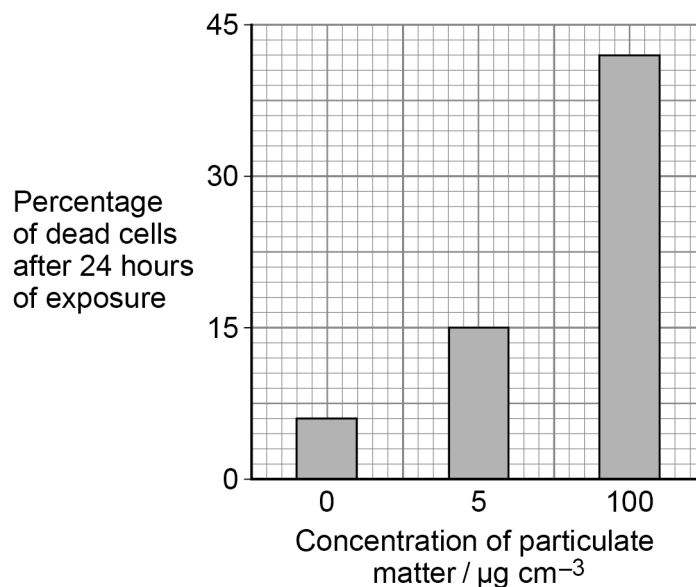
A high concentration of particulate matter results in the death of some alveolar epithelium cells. If alveolar epithelium cells die inside the human body they are replaced by non-specialised, thickened tissue.

Explain why death of alveolar epithelium cells reduces gas exchange in human lungs.

[3 marks]

Scientists grew alveolar epithelium cells and exposed the epithelium cells to different concentrations of particulate matter. They calculated the percentage of these alveolar epithelium cells that died after 24 hours of exposure to particulate matter. Their results are shown in **Figure 6**.

Figure 6



0 6 . 2

Do the data in **Figure 6** show a linear relationship between concentration of particulate matter and percentage of dead cells?

Use suitable calculations to justify your answer.

[2 marks]

Space for your calculations:
